1 A8. Netflix Insights and Handoff

Full version with statistics in writeup.pdf; summary presented here.

1.1 5 Key Insights

Insight 1: Movies Dominate, but TV Shows Are Growing

Movies account for approximately 69.6% of all Netflix titles. However, the share of TV Shows among newly added content has increased from around 22% in 2016 to 35–38% in 2019–2020. This suggests a growing focus on serialized content, although the dataset does not directly reveal Netflix's production decisions.

Insight 2: Recent Content with a Deep Historical Library

The median release year is approximately 2017. However, around 251 titles were released before 1990, and approximately 15% of all titles were added to Netflix more than 10 years after their original release. This shows that Netflix maintains both recent content and a collection of classic films.

Insight 3: Mature Content Dominates

TV-MA is the most common rating, accounting for approximately 36% of all rated titles. For Movies, R and PG-13 ratings together represent around 40% of content. This indicates a substantial proportion of mature content, although family-friendly titles are also available.

Insight 4: South Korean TV Shows Have the Shortest Addition Gap

Among the five leading content-producing countries, South Korean TV Shows have a median release-to-Netflix addition gap of approximately 0–1 year. In comparison, the gap is around two years for the United States and three years for India. This is consistent with Netflix adding Korean dramas relatively soon after release, although the dataset cannot confirm licensing or exclusivity arrangements.

Insight 5: Genre Profiles Differ Across Countries

Netflix's content profiles vary by country. Anime is strongly associated with Japan, K-dramas with South Korea, and Stand-Up Comedy is prominent in the United States. The *International Movies* category can inflate international genre counts, so it should not be treated as a conventional genre in further analysis.

1.2 Handoff Note for Assignment 3

1.2.1 1. Most Informative Variables

The following variables are suitable for feature selection:

- `type`: Distinguishes Movies from TV Shows.
- `release_year`: Indicates the original release year.
- `year_added`: Identifies when content was added to Netflix.
- `rating`: Represents the content rating.
- `duration_int`: Stores duration as a numeric value.
- `num_genres`: Counts the genres associated with a title.
- `num_countries`: Counts the countries associated with a title.
- `gap_years`: Measures the gap between the original release and Netflix addition.

1.2.2 2. Redundant or Overlapping Variables

- `duration`: Use `duration_int` instead because it contains numeric duration values.
- `date_added`: Use `year_added` and `month_added` when analyzing specific time patterns.
- `month_added`: May have limited predictive value unless the analysis focuses on seasonal trends.

1.2.3 3. Variables to Exclude from Features

- `show_id`: An identifier that does not provide meaningful predictive information.
- `title`: Free text that would require NLP techniques to analyze.
- `description`: Free text that would require text preprocessing and NLP.
- `cast`: Has high cardinality, and many entries are missing or marked as `Unknown`.
- `director`: Has high cardinality, and many entries are missing or marked as `Unknown`.

1.2.4 4. Data Cleaning Decisions

- Three rows had duration values in the `rating` column. These were corrected.
- Missing values in `director`, `cast`, and `country` were filled with `Unknown`. This value should be treated as a separate category rather than an actual director, cast member, or country.
- The `gap_years` column contains some negative values due to date inconsistencies. For Assignment 3, filter the dataset to retain only records where `gap_years` ≥ 0.

2 B8. Supermarket Sales Insights and Handoff

Full version with statistics in writeup.pdf; summary presented here.

2.1 5 Key Insights

Insight 1: The Dataset Has Significant Built-In Redundancy

`Tax 5%`, `Cost of goods sold`, `Gross income`, and `Gross margin percentage` are all fully derivable from `Unit price` and `Quantity`. The `Gross margin percentage` is a constant (4.761905%) across all 1000 rows. These columns carry zero independent information. Using them as features in Assignment 3 would introduce artificial correlations and must be avoided.

Insight 2: Food and Beverages Leads Total Revenue, but Home and Lifestyle Has the Highest Average Transaction Value

Food and Beverages generated approximately \$56,145 in total revenue, but its mean transaction value is approximately \$323. Home and Lifestyle has the highest mean transaction value (approximately \$337), meaning customers buying from that category spend more per trip on average. The ranking shifts depending on which metric is used, total revenue or average value, which matters for different business decisions.

Insight 3: Transaction Volume Peaks at 10 AM, 1 PM, and 7 PM

These three hours account for a disproportionate share of daily transactions, forming a clear three-hump pattern consistent with morning shopping, lunch break shopping, and post-work shopping. Saturday has the highest transaction volume (approximately 164 transactions), while Monday is the quietest (approximately 125 transactions). Mean transaction value, however, does not change meaningfully across hours or days.

Insight 4: Branch, Customer Type, and Payment Method Show Minimal Effect on Spending

Mean transaction values across Branch A (\$312), B (\$319), and C (\$318) differ by less than \$7. Members and Normal customers spend within \$5 of each other per transaction. The three payment methods also show nearly identical mean totals. All differences are small relative to the within-group standard deviation (approximately \$181). These variables are not strong drivers of transaction value.

Insight 5: Customer Rating Is Uncorrelated with All Transactional Variables

Pearson correlation between rating and unit price is approximately 0.01 ($p > 0.05$), and between rating and total spend is approximately 0.01 ($p > 0.05$). Rating distributions are nearly identical across all product lines and branches. This is a meaningful null result: whatever drives customer satisfaction, it is not captured by any other variable in this

dataset. Rating should be treated as an outcome variable, not a predictor, and its weak relationships with other features should be disclosed in any modelling work.

2.2 Handoff Note for Assignment 3

Maximum 150 words.

Most Informative Variables:

`Unit price`, `Quantity`, `Total` (pick one as target or use for regression), `Branch`, `Product line`, `Customer type`, `Gender`, `Payment`, `Hour`, `Weekday`, and `Month`.

Redundant Variables: Do Not Use as Independent Features

`Tax 5%`, `Cost of goods sold`, and `Gross income` are all functions of `Unit price` and `Quantity`. `Gross margin percentage` is constant and has zero variance. `City` is identical to `Branch`. Use derived variables such as `Hour`, `Weekday`, and `Month` instead of the original `Date` and `Time` columns.

Variable Not to Treat as an Analytical Feature

`Invoice ID` is an identifier and should not be used as a predictive feature.

Cleaning Decisions to Carry Forward

No missing values or duplicates were found. The nine high-`Total` outliers (Quantity = 10, high unit price) are valid transactions and were retained. `Customer stratification rating` is nearly uncorrelated with all other variables. If used as a target, expect low predictive accuracy from the available features. Branch and City are interchangeable; choose one.